

COMPUTER PROGRAMMING ASSIGNMENT

ICS2102 / HBC2204 / HCB2351

C PROGRAMMING ASSIGNMENT

NAME: Ekirapa Japheth

REGISTRATION NUMBER: HDB212-D1-0023/2021

COURSE: BBIT

SECTIONS COVERED:

- Section A – Arrays
- Section B – Structures and Functions
- Section C – Pointers and Dynamic Memory Allocation

SECTION A: ARRAYS

Problem Statement

Develop a C program that stores the marks of 10 students in an array. The program should read the marks using a loop, calculate the average mark, determine the highest and lowest marks, and display all marks that are greater than the average.

Algorithm

1. Start the program.
2. Declare an array to store marks of 10 students.
3. Use a loop to input the marks of the 10 students.
4. Add all the marks to calculate the total.
5. Calculate the average by dividing the total by 10.
6. Set the first mark as the initial highest and lowest mark.
7. Compare the remaining marks to determine the highest and lowest marks.
8. Display the average, highest mark, and lowest mark.
9. Compare each mark with the average and display marks greater than the average.
10. End the program.

Source Code:

```
#include <stdio.h>

int main()
{
    int marks[10];

    int i;

    int highest, lowest;

    float average, total = 0;

    printf("=====\n");

    printf("        SECTION A - ARRAYS\n");

    printf("=====\n");

    printf("Name: Ekirapa Japheth\n");
```

```
printf("Registration Number: HDB212-D1-0023/2021\n");
```

```
printf("Course: BBIT\n\n");
```

```
printf("Enter marks for 10 students:\n");
```

```
for(i = 0; i < 10; i++)
```

```
{
```

```
    printf("Enter mark for student %d: ", i + 1);
```

```
    scanf("%d", &marks[i]);
```

```
    total = total + marks[i];
```

```
}
```

```
average = total / 10;
```

```
highest = marks[0];
```

```
lowest = marks[0];
```

```
for(i = 1; i < 10; i++)
```

```
{
```

```
    if(marks[i] > highest)
```

```
    {
```

```
        highest = marks[i];
```

```
    }
```

```
    if(marks[i] < lowest)
```

```
    {
```

```

        lowest = marks[i];

    }

}

printf("\n=====\\n");

printf("                RESULTS\\n");

printf("=====\\n");


printf("Average mark: %.2f\\n", average);

printf("Highest mark: %d\\n", highest);

printf("Lowest mark: %d\\n", lowest);


printf("\\nMarks greater than the average:\\n");


for(i = 0; i < 10; i++)
{
    if(marks[i] > average)
    {
        printf("%d ", marks[i]);
    }
}


printf("\\n\\nProgram completed successfully.\\n");

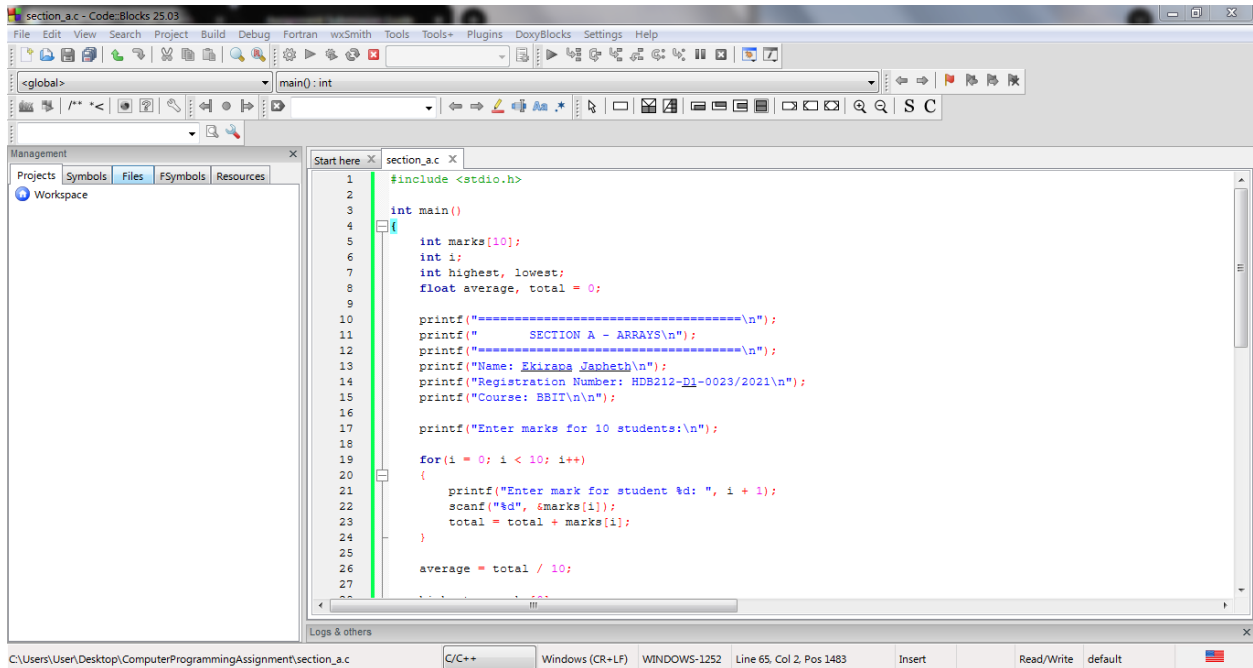

return 0;

}

```

Screenshots

Section A: Source code – top part

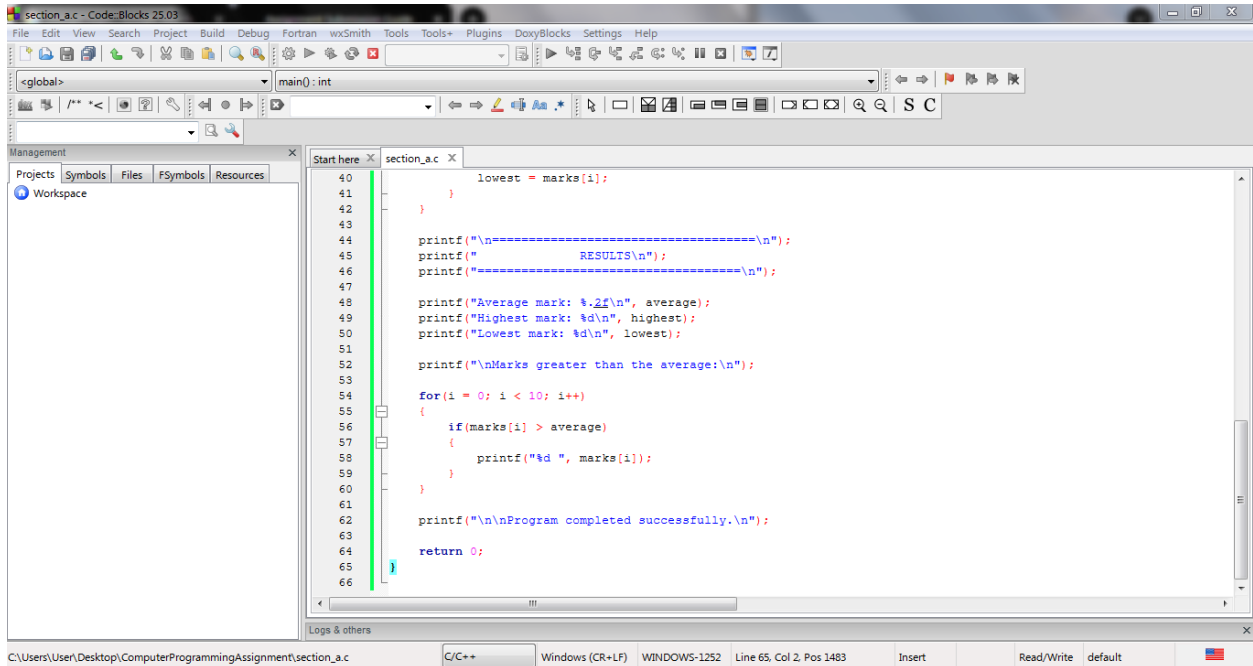


The screenshot shows the Code::Blocks IDE with the file 'section_a.c' open. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int marks[10];
6     int i;
7     int highest, lowest;
8     float average, total = 0;
9
10    printf("=====\\n");
11    printf("SECTION A - ARRAYS\\n");
12    printf("=====\\n");
13    printf("Name: Ekirana Japheth\\n");
14    printf("Registration Number: HDB212-D1-0029/2021\\n");
15    printf("Course: BSIT\\n\\n");
16
17    printf("Enter marks for 10 students:\\n");
18
19    for(i = 0; i < 10; i++)
20    {
21        printf("Enter mark for student %d: ", i + 1);
22        scanf("%d", &marks[i]);
23        total = total + marks[i];
24    }
25
26    average = total / 10;
27
```

The status bar at the bottom indicates the file path is 'C:\Users\User\Desktop\ComputerProgrammingAssignment\section_a.c', the language is 'C/C++', and the cursor is at 'Line 65, Col 2, Pos 1483'.

Section A: Source code – bottom part

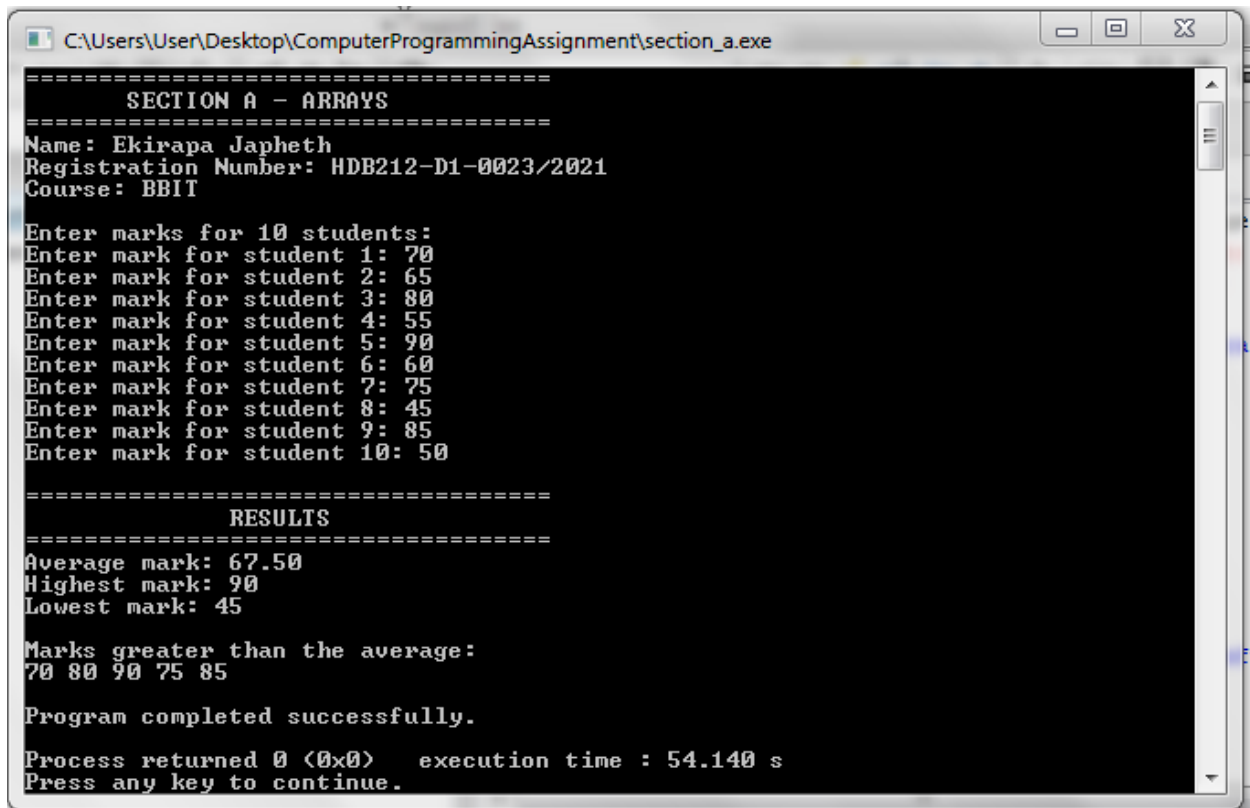


The screenshot shows the Code::Blocks IDE with the file 'section_a.c' open, displaying the bottom part of the code:

```
40         lowest = marks[i];
41     }
42 }
43
44 printf("\\n=====\\n");
45 printf("RESULTS\\n");
46 printf("=====\\n");
47
48 printf("Average mark: %.2f\\n", average);
49 printf("Highest mark: %d\\n", highest);
50 printf("Lowest mark: %d\\n", lowest);
51
52 printf("\\nMarks greater than the average:\\n");
53
54 for(i = 0; i < 10; i++)
55 {
56     if(marks[i] > average)
57     {
58         printf("%d ", marks[i]);
59     }
60 }
61
62 printf("\\n\\nProgram completed successfully.\\n");
63
64 return 0;
65
66
```

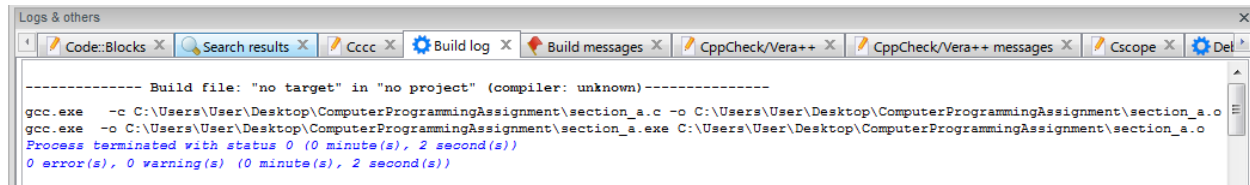
The status bar at the bottom indicates the file path is 'C:\Users\User\Desktop\ComputerProgrammingAssignment\section_a.c', the language is 'C/C++', and the cursor is at 'Line 65, Col 2, Pos 1483'.

Section A: Program execution



```
=====  
SECTION A - ARRAYS  
=====  
Name: Ekirapa Japheth  
Registration Number: HDB212-D1-0023/2021  
Course: BBIT  
  
Enter marks for 10 students:  
Enter mark for student 1: 70  
Enter mark for student 2: 65  
Enter mark for student 3: 80  
Enter mark for student 4: 55  
Enter mark for student 5: 90  
Enter mark for student 6: 60  
Enter mark for student 7: 75  
Enter mark for student 8: 45  
Enter mark for student 9: 85  
Enter mark for student 10: 50  
  
=====  
RESULTS  
=====  
Average mark: 67.50  
Highest mark: 90  
Lowest mark: 45  
  
Marks greater than the average:  
70 80 90 75 85  
  
Program completed successfully.  
  
Process returned 0 (0x0) execution time : 54.140 s  
Press any key to continue.
```

Section A: Successful compilation



```
----- Build file: "no target" in "no project" (compiler: unknown) -----  
gcc.exe -c C:\Users\User\Desktop\ComputerProgrammingAssignment\section_a.c -o C:\Users\User\Desktop\ComputerProgrammingAssignment\section_a.o  
gcc.exe -o C:\Users\User\Desktop\ComputerProgrammingAssignment\section_a.exe C:\Users\User\Desktop\ComputerProgrammingAssignment\section_a.o  
Process terminated with status 0 (0 minute(s), 2 second(s))  
0 error(s), 0 warning(s) (0 minute(s), 2 second(s))
```

SECTION B: STRUCTURES AND FUNCTIONS

Problem Statement

Develop a Student Record Management Program using structures and functions. The program should store information for five students, including Admission Number, Student Name, Course, Age, and CAT Marks. The program should capture and display student information and calculate the average CAT marks.

Algorithm

1. Start the program.
2. Define a structure called `Student`.
3. Create variables to store admission number, name, course, age, and CAT marks.
4. Declare an array to store records for 5 students.
5. Use a function to capture information for each student.
6. Use a function to display the information of each student.
7. Use a function to calculate the average CAT marks.
8. Display the average CAT marks.
9. End the program.

Source Code:

```
#include <stdio.h>

struct Student
{
    char admissionNumber[30];
    char name[50];
    char course[50];
    int age;
    float catMarks;
};

void captureStudent(struct Student *student, int number)
{
    printf("\nEnter information for Student %d\n", number);

    printf("Admission Number: ");
    scanf("%[^\n]", student->admissionNumber);
```

```

printf("Student Name: ");
scanf("%[^\n]", student->name);

printf("Course: ");
scanf("%[^\n]", student->course);

printf("Age: ");
scanf("%d", &student->age);

printf("CAT Marks: ");
scanf("%f", &student->catMarks);
}

void displayStudent(struct Student student, int number)
{
    printf("\n-----\n");
    printf("Student %d\n", number);
    printf("-----\n");
    printf("Admission Number: %s\n", student.admissionNumber);
    printf("Student Name: %s\n", student.name);
    printf("Course: %s\n", student.course);
    printf("Age: %d\n", student.age);
    printf("CAT Marks: %.2f\n", student.catMarks);
}

float calculateAverage(struct Student students[], int size)
{
    float total = 0;
    int i;

    for(i = 0; i < size; i++)
    {

```



```

        total = total + students[i].catMarks;
    }

    return total / size;
}

int main()
{
    struct Student students[5];
    float average;
    int i;

    printf("=====\n");
    printf("        SECTION B - STUDENT RECORD\n");
    printf("=====\n");
    printf("Name: Ekirapa Japheth\n");
    printf("Registration Number: HDB212-D1-0023/2021\n");
    printf("Course: BBIT\n");

    for(i = 0; i < 5; i++)
    {
        captureStudent(&students[i], i + 1);
    }

    printf("\n\n=====\n");
    printf("        STUDENT RECORDS\n");
    printf("=====\n");

    for(i = 0; i < 5; i++)
    {
        displayStudent(students[i], i + 1);
    }
}

```

```

        average = calculateAverage(students, 5);

        printf("\n=====\\n");

        printf("Average CAT Marks: %.2f\\n", average);

        printf("=====\\n");

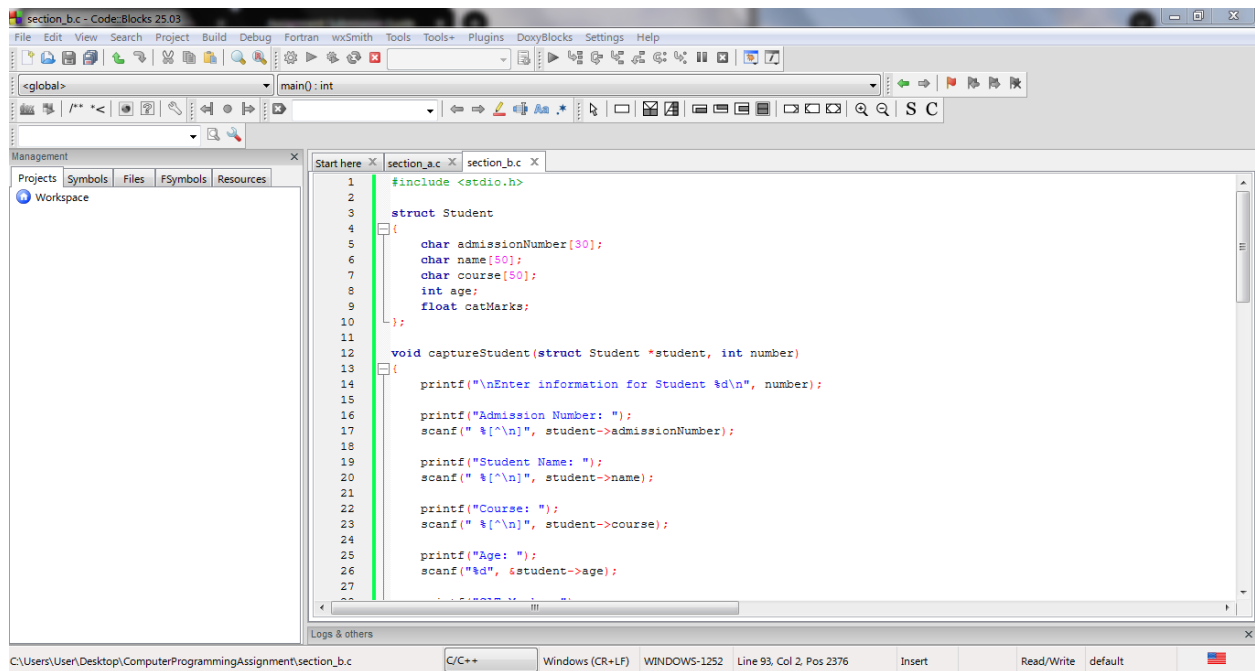
        printf("\\nProgram completed successfully.\\n");

        return 0;
}

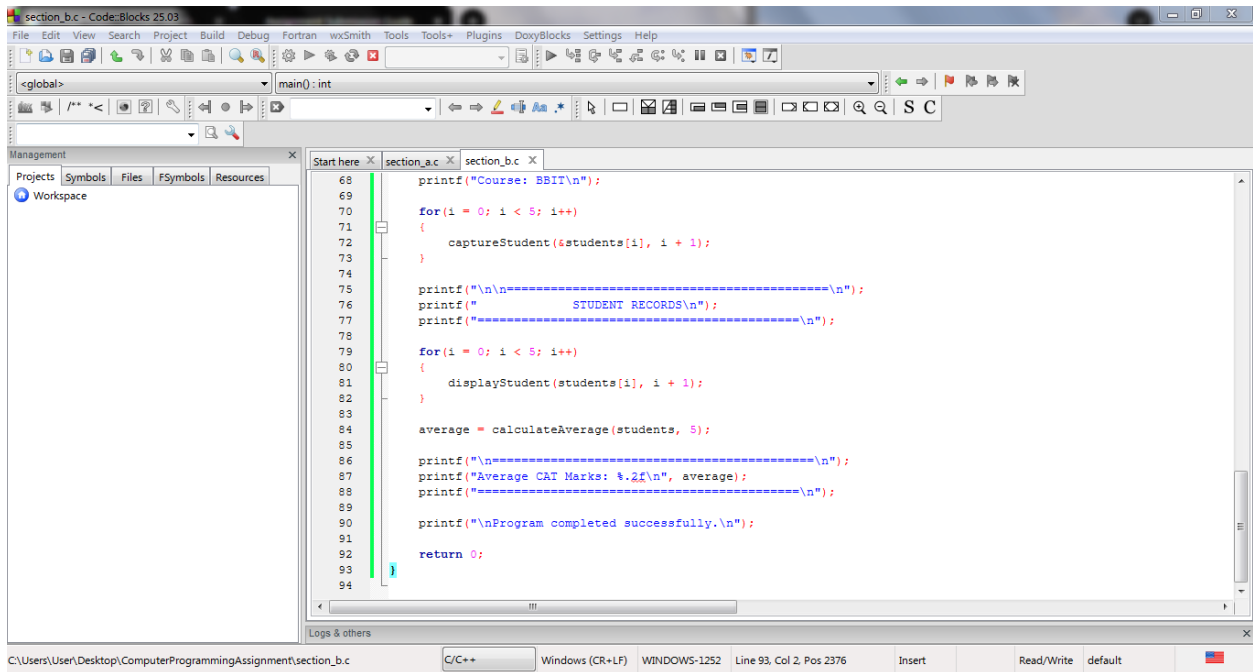
```

Screenshots

Section B: Source code – top part

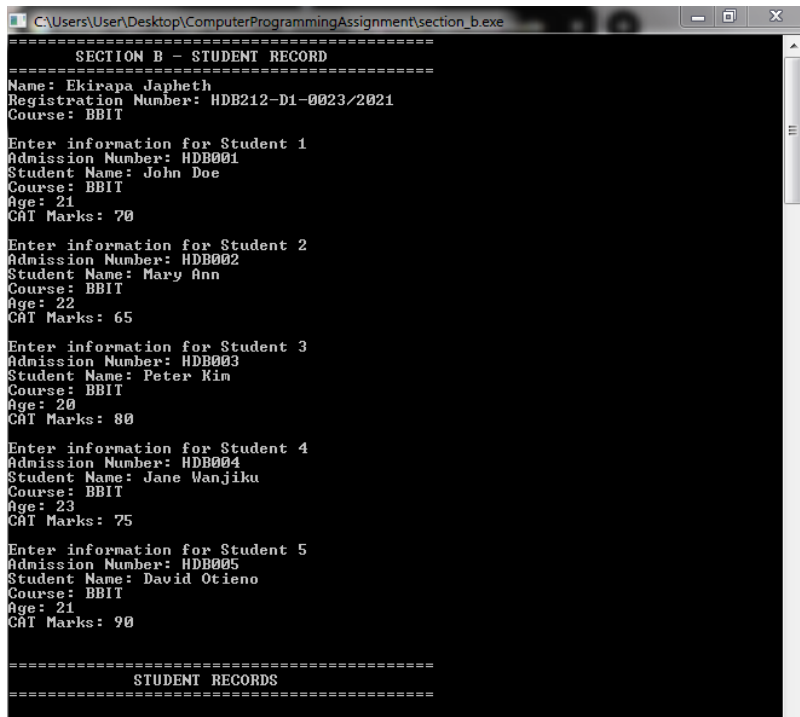


Section B: Source code – bottom part



```
68     printf("Course: BBIT\n");
69
70     for(i = 0; i < 5; i++)
71     {
72         captureStudent(&students[i], i + 1);
73     }
74
75     printf("\n\n===== \n");
76     printf("                STUDENT RECORDS\n");
77     printf("===== \n");
78
79     for(i = 0; i < 5; i++)
80     {
81         displayStudent(students[i], i + 1);
82     }
83
84     average = calculateAverage(students, 5);
85
86     printf("\n===== \n");
87     printf("Average CAT Marks: %.2f\n", average);
88     printf("===== \n");
89
90     printf("\nProgram completed successfully.\n");
91
92     return 0;
93
94
```

Section B program execution



```
=====
SECTION B - STUDENT RECORD
=====
Name: Ekirapa Japheth
Registration Number: HDB212-D1-0023/2021
Course: BBIT

Enter information for Student 1
Admission Number: HDB001
Student Name: John Doe
Course: BBIT
Age: 21
CAT Marks: 70

Enter information for Student 2
Admission Number: HDB002
Student Name: Mary Ann
Course: BBIT
Age: 22
CAT Marks: 65

Enter information for Student 3
Admission Number: HDB003
Student Name: Peter Kim
Course: BBIT
Age: 20
CAT Marks: 80

Enter information for Student 4
Admission Number: HDB004
Student Name: Jane Wanjiku
Course: BBIT
Age: 23
CAT Marks: 75

Enter information for Student 5
Admission Number: HDB005
Student Name: David Otieno
Course: BBIT
Age: 21
CAT Marks: 90

=====
STUDENT RECORDS
=====
```

```
C:\Users\User\Desktop\ComputerProgrammingAssignment\section_b.exe

=====
STUDENT RECORDS
=====

Student 1
-----
Admission Number: HDB001
Student Name: John Doe
Course: BBIT
Age: 21
CAT Marks: 70.00

Student 2
-----
Admission Number: HDB002
Student Name: Mary Ann
Course: BBIT
Age: 22
CAT Marks: 65.00

Student 3
-----
Admission Number: HDB003
Student Name: Peter Kim
Course: BBIT
Age: 20
CAT Marks: 80.00

Student 4
-----
Admission Number: HDB004
Student Name: Jane Wanjiku
Course: BBIT
Age: 23
CAT Marks: 75.00

Student 5
-----
Admission Number: HDB005
Student Name: David Otieno
Course: BBIT
Age: 21
CAT Marks: 90.00

=====
Average CAT Marks: 76.00
=====

Program completed successfully.
Process returned 0 (0x0)   execution time : 345.335 s
```

Section B: Successful compilation

```
Logs & others
Code::Blocks Search results Cccc Build log Build messages CppCheck/Vera++ CppCheck/Vera++ messages Cscope Det

----- Build file: "no target" in "no project" (compiler: unknown)-----

gcc.exe -c C:\Users\User\Desktop\ComputerProgrammingAssignment\section_b.c -o C:\Users\User\Desktop\ComputerProgrammingAssignment\section_b.o
gcc.exe -o C:\Users\User\Desktop\ComputerProgrammingAssignment\section_b.exe C:\Users\User\Desktop\ComputerProgrammingAssignment\section_b.o
Process terminated with status 0 (0 minute(s), 0 second(s))
0 error(s), 0 warning(s) (0 minute(s), 0 second(s))

ection_b.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 93, Col 2, Pos 2376 Insert Read/Write default
```

SECTION C: POINTERS AND DYNAMIC MEMORY ALLOCATION

Problem Statement

Develop a C program that dynamically allocates memory for 15 integers using `malloc()`. The program should allow the user to enter 15 integers, calculate their total and average, and free the allocated memory before the program terminates.

Algorithm

1. Start the program.
2. Declare a pointer to an integer.
3. Dynamically allocate memory for 15 integers using `malloc()`.
4. Check whether memory allocation was successful.
5. Use a loop to read 15 integers.
6. Add the integers to calculate the total.
7. Calculate the average of the 15 integers.
8. Display the total and average.
9. Free the allocated memory using `free()`.
10. Display a message confirming that the memory has been freed.
11. End the program.

Source Code:

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int *numbers;

    int i;

    int total = 0;

    float average;

    printf("=====\n");

    printf(" SECTION C - POINTERS AND DYNAMIC MEMORY\n");

    printf("=====\n");
```

```

printf("Name: Ekirapa Japheth\n");

printf("Registration Number: HDB212-D1-0023/2021\n");

printf("Course: BBIT\n\n");


numbers = (int *)malloc(15 * sizeof(int));


if(numbers == NULL)

{

    printf("Memory allocation failed.\n");

    return 1;

}


printf("Enter 15 integers:\n");


for(i = 0; i < 15; i++)

{

    printf("Enter number %d: ", i + 1);

    scanf("%d", &numbers[i]);

    total = total + numbers[i];

}


average = (float)total / 15;


printf("\n===== \n");

printf("                RESULTS\n");

printf("===== \n");

```

```

printf("Total: %d\n", total);

printf("Average: %.2f\n", average);


free(numbers);


printf("\nMemory has been freed successfully.\n");

printf("Program completed successfully.\n");

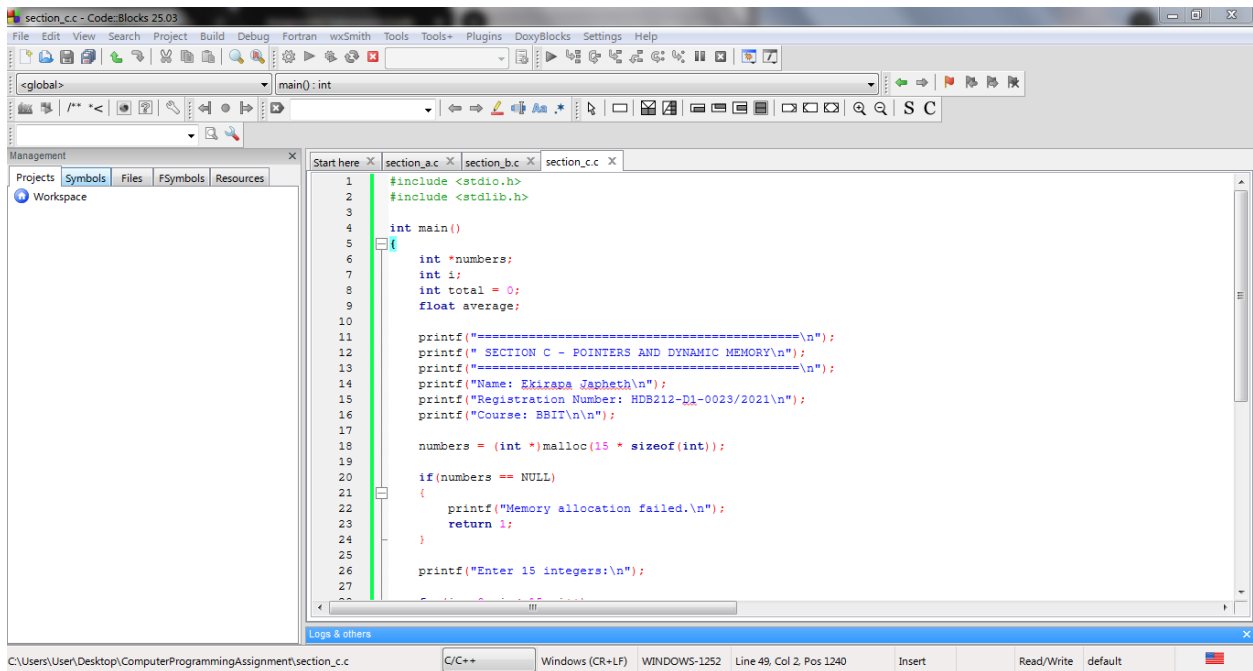

return 0;

}

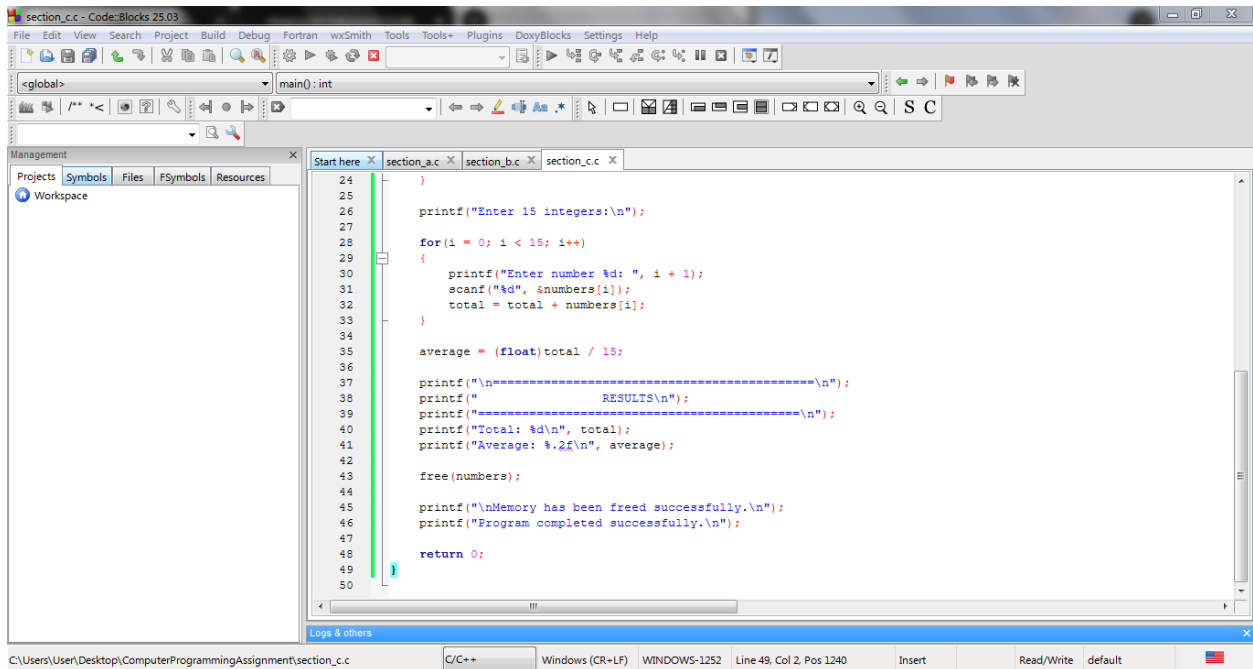
```

Screenshots

Section C: Source code – top part



Section C: Source code – bottom part

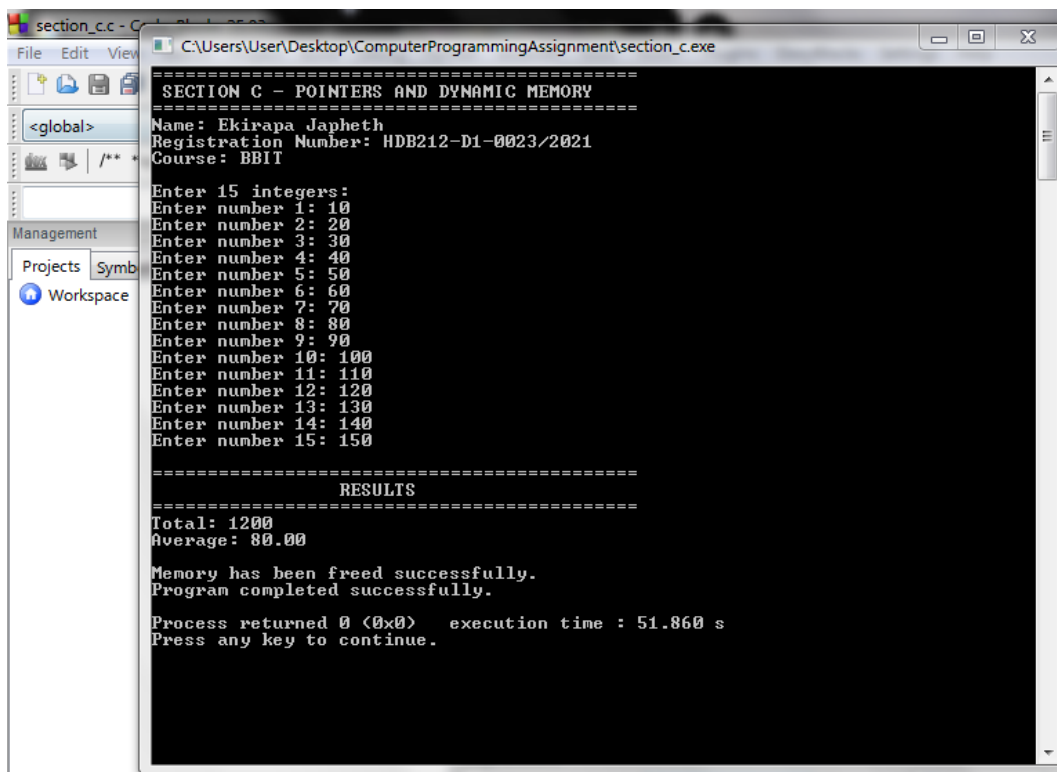


```
24     }
25
26     printf("Enter 15 integers:\n");
27
28     for(i = 0; i < 15; i++)
29     {
30         printf("Enter number %d: ", i + 1);
31         scanf("%d", &numbers[i]);
32         total = total + numbers[i];
33     }
34
35     average = (float)total / 15;
36
37     printf("\n===== \n");
38     printf("                RESULTS\n");
39     printf("===== \n");
40     printf("Total: %d\n", total);
41     printf("Average: %.2f\n", average);
42
43     free(numbers);
44
45     printf("\nMemory has been freed successfully.\n");
46     printf("Program completed successfully.\n");
47
48     return 0;
49 }
50
```

Logs & others

C:\Users\User\Desktop\ComputerProgrammingAssignment\section_c.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 49, Col 2, Pos 1240 Insert Read/Write default

Section C program execution



```
SECTION C - POINTERS AND DYNAMIC MEMORY
=====
Name: Ekirapa Japheth
Registration Number: HDB212-D1-0023/2021
Course: BBIT

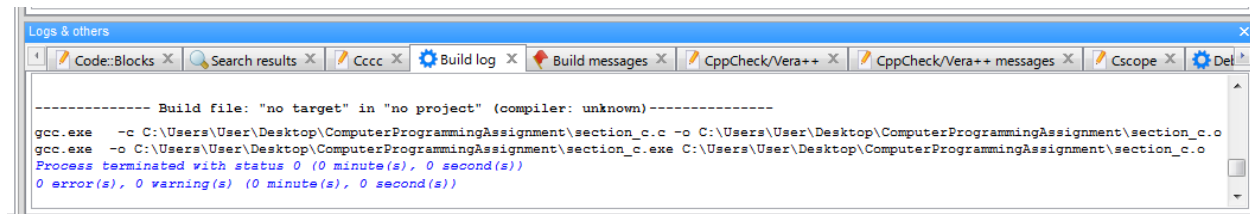
Enter 15 integers:
Enter number 1: 10
Enter number 2: 20
Enter number 3: 30
Enter number 4: 40
Enter number 5: 50
Enter number 6: 60
Enter number 7: 70
Enter number 8: 80
Enter number 9: 90
Enter number 10: 100
Enter number 11: 110
Enter number 12: 120
Enter number 13: 130
Enter number 14: 140
Enter number 15: 150

=====
RESULTS
=====
Total: 1200
Average: 80.00

Memory has been freed successfully.
Program completed successfully.

Process returned 0 (0x0)   execution time : 51.860 s
Press any key to continue.
```


Section C: Successful compilation



```
----- Build file: "no target" in "no project" (compiler: unknown)-----
gcc.exe -c C:\Users\User\Desktop\ComputerProgrammingAssignment\section_c.c -o C:\Users\User\Desktop\ComputerProgrammingAssignment\section_c.o
gcc.exe -o C:\Users\User\Desktop\ComputerProgrammingAssignment\section_c.exe C:\Users\User\Desktop\ComputerProgrammingAssignment\section_c.o
Process terminated with status 0 (0 minute(s), 0 second(s))
0 error(s), 0 warning(s) (0 minute(s), 0 second(s))
```

CONCLUSION

The assignment provided practical experience in C programming through the use of arrays, structures, functions, pointers, and dynamic memory allocation. Section A demonstrated how arrays and loops can be used to process student marks. Section B demonstrated the use of structures and functions to manage student records and calculate CAT marks. Section C demonstrated dynamic memory allocation using `malloc()` and the release of allocated memory using `free()`. The programs were compiled and executed successfully.